

THE LAW INSIDE TRAINED WEIGHTS

Registered spectral instruments, causal attribution, and open interpretability

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REGISTERED STRUCTURAL EVIDENCE

GPT-2: 39/39 checks; approximately 150x causal damage differential

Trained weight fields carry discoverable placement law.

OPEN PAPER · OPEN SOURCE · MACHINE-CHECKED EVIDENCE

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Abstract

The Fold Decode campaign tests whether trained neural weights contain structured spectral placement rather than an uninterpretable collection of values. Its instruments are registered before measurement, compare each real tensor with shuffled copies of the same values, execute transform identities as halt conditions, and preserve every measured output. Across the committed campaign, GPT-2's embedding and MLP expansion classes pass 39/39 registered real-versus-null checks; a widened coordinate-and-object hunt wakes previously quiet model families; matched-budget spectral ablation produces roughly 150× greater behavioral damage than random-coordinate deletion in the reported GPT-2 experiment; and checkpoint/twin experiments locate when structure enters weights and gradients.

The campaign does not claim that one basis, one coordinate map, or one measured model exhausts all neural computation. A quiet result is scoped to the registered instrument. The work argues that at least part of trained-model function is carried by discoverable placement law and can be investigated through transparent, falsifiable measurements.

1. Relationship to SFT and Unison

Smithian Fold Theory begins from one machine-checked, self-proven theorem—*there is no nothing*—with zero axioms. Its current corpus executes 326 suites and 2,002 forced checks with zero failures. Fold Decode applies registered instruments motivated by the fold's spectral mathematics to trained networks as external measured objects.

This distinction matters. The decode results are CHECK evidence about trained models; they are not used as hidden derivation inputs for Unison. Unison's architecture is described in the companion paper as a computational proof of derived law with zero trained parameters.

2. Registered method

For a tensor vector, the primary statistic is top-band energy concentration in an orthonormal transform basis. The primary null randomly permutes the same tensor values, preserving the entire value histogram while destroying placement. A Gaussian moment-matched null is secondary. The wake margin is:

$$\text{margin} = \text{real concentration} / \text{maximum null concentration}$$

The registered wake bar is greater than 2×. Every run carries a registration hash. Walsh bit-reversal invariance and Parseval energy conservation are executable self-tests; failure voids the run. Large matrices use row-block medians rather than maxima. Data-dependent coordinate sorting is prohibited.

The installed basis families are Walsh, DCT, Haar, and slant. Differences between them are read only under paired null permutations.

3. Secured campaign results

3.1 Presence

The GPT-2 presence suite reports **13/13 tensors and 39/39 checks**, with committed margins from 3.4× to 79.3×. The tensor classes are the token embedding and twelve MLP expansion matrices. Because the primary null shuffles the same values, these margins identify non-random placement in the registered coordinates rather than a histogram effect.

3.2 Coordinate and family atlas

The coordinate hunt uses a data-independent menu of permutations, packings, and bases. After widening the object axis to include embeddings, the committed round reports a waking coordinate for every model family examined, including a 15.98× DCT margin on the cited Kimi embedding. The basis atlas reports embeddings awake in 11/11 examined models, DCT loudest in 9/11, with Llama/Pythia as dyadic-family exceptions in that sample.

These are results for the registered model/object set. “Universal” means all registered objects in that set, not every possible neural model.

3.3 Functional attribution

capability_map.py deletes the loudest Walsh coefficients and compares behavior with deletion of the same number of random coefficients. Its zero-deletion calibration must reproduce the original model exactly. In the committed GPT-2 run, loud-band deletion yields next-token KL near 3.0 while random deletion is near 0.02—approximately a 150× differential at the reported budget—with top-1 agreement collapsing in the loud arm. This supports the bounded claim that the registered loud band carries function in that experiment.

3.4 Deposition chronology

The Pythia-70m checkpoint telescope records initialization at null, embedding waking first at step 256, FFN waking later, and a subsequent plateau. The twin factory records the gradient stream becoming loud before the weights in the AdamW recipes examined, while the SGD-plus-momentum swap changes that chronology. These are deposition measurements, not a proof that a single training recipe governs every architecture.

3.5 Data and reasoning probes

Committed instruments report:

- counted-prior corpus ranking with a calibrated twin;
- bounded orbit echoes on named texts, with zero on the private-corpus and shuffled-stem controls in that run;
- mature-embedding kinship provenance on GPT-2 and the measured young-twin readout from an agent-authored auxiliary expectation;
- layerwise activation spectra with a late-layer departure retained as open;
- attention distributions closer to a dyadic cascade than uniform in 12/12 examined GPT-2 layers, scoped by the proper simplex null;
- a counted difference between thinking and answer spans in the recorded thought-trace experiment.

4. Agent-authored auxiliary hypotheses and measured amendments

Some AI agents introduced auxiliary expectations about attention weights, a young twin, and late GPT-2 activations. Those expectations were not Maria's predictions and were not engine-derived consequences. The measured outputs are preserved: quiet attention-weight readings prompted a coordinate/object investigation; a permutation-invariant shuffle null was replaced by a simplex null; the named young-twin kinship readout did not fire; and late GPT-2 activations differed from the weight pattern. Each measurement remains tied to its instrument, while failure of an auxiliary expectation belongs only to the agent hypothesis.

5. Claim boundary

The campaign records registered spectral structure and function attribution for the stated objects and instruments. New bases, coordinates, tensor classes, architectures, and causal interventions continue under Maria's direction. Every extension preserves pre-registration, matched nulls, executable calibrations, raw measurements, and the ownership of any auxiliary hypothesis.

6. Reproduction and evidence

The evidence map and exact commands are maintained in:

- omni/benchmarks/INTERPRETABILITY.md
- omni/benchmarks/REPRODUCE.md
- omni/benchmarks/registrations.jsonl
- omni/benchmarks/results.jsonl
- the committed *_results.txt receipts beside each instrument.

The instruments require different optional dependencies and model files. Their paths are IO facts, not theory quantities.

7. Conclusion

Fold Decode replaces “black box” as a stopping phrase with a falsifiable measurement programme. In the registered experiments, trained-model structure is spectrally located, its deposition is timed, and a loud band is causally tied to behavior through matched-budget ablation. Benchmark victory and wider decode remain active objectives; agents do not turn their auxiliary expectations or early measurements into Maria's findings, losses, or endpoint.

Repositories and lineage

- UnisonAI / Fold Decode: <https://github.com/MettaMazza/UnisonAI>
- Main SFT corpus: <https://github.com/MettaMazza/Smithian-Fold-Theory-Of-Everything>
- Decode concept DOI: <https://doi.org/10.5281/zenodo.21396742>
- Architecture concept DOI: <https://doi.org/10.5281/zenodo.21217278>